

Smart Traffic Lights Using Local Artificial Intelligence

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Abstract—This paper presents a study of smart traffic-light systems using local artificial intelligence. The proposed approach uses street cameras and local computing devices to analyze traffic conditions and adapt signal timing. Local processing can reduce dependence on cloud communication and support faster traffic decisions. The system architecture, existing approaches, performance measures, and research gap related to emergency vehicle prioritization are discussed.

Index Terms—Artificial Intelligence, Smart Traffic Lights, Edge Computing, Computer Vision, Adaptive Traffic Control, Emergency Vehicles

I. INTRODUCTION

Traditional traffic lights generally operate using fixed timers. Such systems may not respond efficiently to sudden changes in traffic density. Artificial intelligence and computer vision can help estimate vehicle density and adapt signal timing according to road conditions [1], [2].

This paper discusses local AI processing for smart traffic lights and the possibility of prioritizing emergency vehicles.

II. LITERATURE REVIEW

Existing research explores AI-based traffic signal control and adaptive signal timing. Rasheed et al. reviewed deep reinforcement learning approaches for traffic signal control [1]. Gorodokin et al. investigated the optimization of adaptive traffic-light control modes using machine vision [2].

Cloud-based AI systems send camera information to a remote server for processing. This can provide access to substantial computing resources, but it depends on internet connectivity and may introduce communication delay.

Local smart-camera systems move some processing closer to the road. This can reduce the need to transmit all camera data over the internet. However, hardware capabilities, processing delay, and system cost must be considered.

The supplied literature survey also lists a 2024 paper attributed to J. Smith et al. on real-time traffic processing using low-cost microcontrollers. Its publication details and reported claims should be verified before being treated as established evidence [3].

III. PROPOSED METHODOLOGY

The proposed system follows these steps:

- 1) Capture road conditions using street cameras.
- 2) Process camera information using local AI.

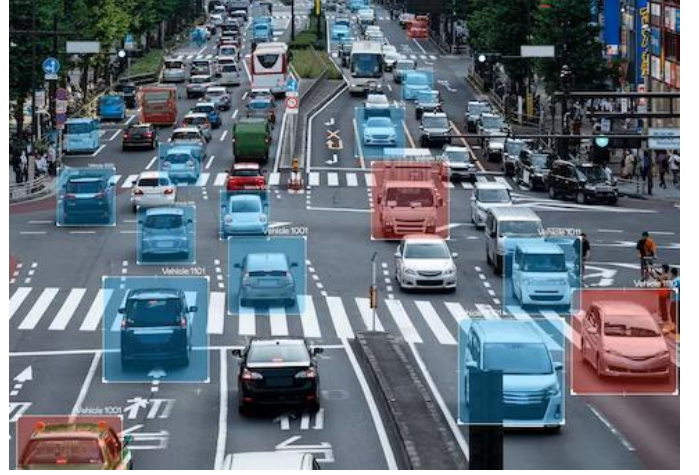


Fig. 1. AI-based vehicle detection using road cameras.

- 3) Estimate vehicle density and identify emergency vehicles.
- 4) Determine appropriate traffic-signal timing.
- 5) Send control instructions to the traffic lights.
- 6) Forward selected information to the cloud for storage and analysis.

IV. SYSTEM ARCHITECTURE

V. SYSTEM FLOWCHART

VI. PERFORMANCE EVALUATION

The average vehicle waiting time can be expressed as

$$T_{avg} = \frac{1}{N} \sum_{i=1}^N T_i \quad (1)$$

where N is the number of vehicles and T_i is the waiting time of vehicle i .

The percentage reduction in waiting time is

$$R = \frac{T_{base} - T_{AI}}{T_{base}} \times 100 \quad (2)$$

where T_{base} is the baseline waiting time and T_{AI} is the waiting time under AI-based control.

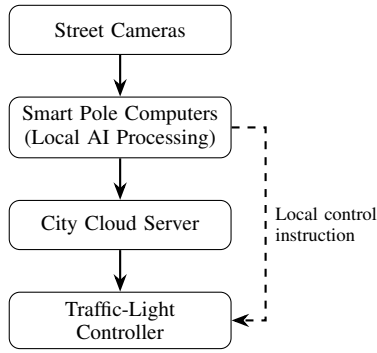


Fig. 2. Architecture of the local AI-based traffic system.

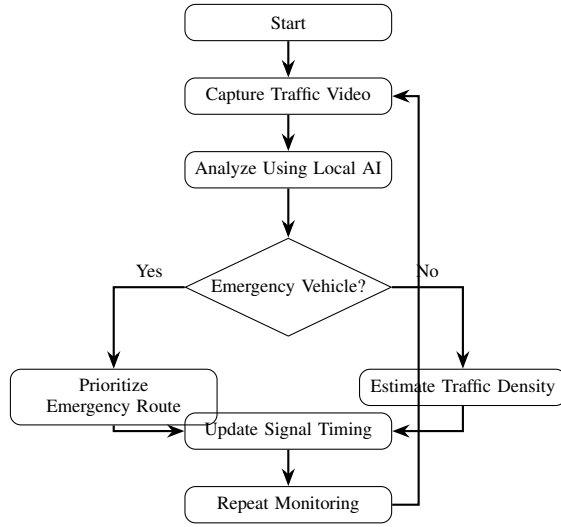


Fig. 3. Flowchart of the proposed traffic-control process.

VII. COMPARISON OF TRAFFIC SYSTEMS

Table ?? compares the approaches using the values provided in the supplied literature survey.

VIII. RESEARCH GAP

Existing traffic-control systems can handle predictable traffic conditions. However, automatic detection of emergency vehicles and coordination of nearby traffic signals remain important areas for further investigation.

The proposed approach considers emergency vehicle prioritization as a potential extension to local AI-based traffic control.

IX. RESULTS AND DISCUSSION

The system can be evaluated using average waiting time, decision latency, internet data usage, and setup cost. Experimental testing is required to establish whether local AI processing improves these measures under different traffic conditions.

X. FUTURE SCOPE

Future work may include emergency vehicle detection, coordination between multiple intersections, improved vehicle classification, and testing with real-time traffic data. Hardware performance and system reliability should also be evaluated.

XI. CONCLUSION

Local AI-based traffic control provides an approach to adapting traffic signals to changing road conditions. Combining camera-based vehicle detection with local processing may reduce dependence on cloud communication. Further implementation and testing are needed to evaluate performance and emergency vehicle prioritization.

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